

SCIENCE 2025 ALL SETS

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CHAPTER: 1 - Chemical Reactions and Equations (Chemistry)

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YEAR: 2025

CHAPTER: 1 - Chemical Reactions and Equations

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: In which one of the following situations a chemical reaction does not occur?

OPTIONS: (A) Milk is left open at room temperature during summer (B) Grapes get fermented (C) An iron nail is left exposed to humid atmosphere (D) Melting of glaciers

ANSWER: (D) Melting of glaciers

YEAR: 2025

CHAPTER: 1 - Chemical Reactions and Equations

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: In order to prepare dry hydrogen chloride gas in humid atmosphere, the gas produced is passed through a guard tube (drying tube) which contains:

OPTIONS: (A) Calcium chloride (B) Calcium oxide (C) Calcium hydroxide (D) Calcium carbonate

ANSWER: (A) Calcium chloride

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CHAPTER: 1 - Chemical Reactions and Equations

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The property by virtue of which a solid material can be drawn into thin wires is called:

OPTIONS: (A) malleability (B) ductility (C) rigidity (D) resistivity

ANSWER: (B) ductility

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CHAPTER: 1 - Chemical Reactions and Equations

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Select from the following a hydrocarbon having one C-C bond and one C=C bond:

OPTIONS: (A) Benzene (B) Cyclohexane (C) Butyne (D) Propyne

ANSWER: (D) Propyne

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CHAPTER: 1 - Chemical Reactions and Equations

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The main observations while performing the experiment of burning magnesium ribbon in air are: (i) Magnesium ribbon burns with a dazzling white flame. (ii) A white powder is formed. (iii) Magnesium ribbon vapourises. (iv) Aqueous solution of the white powder turns blue litmus to red.

OPTIONS: (A) (i) and (iv) (B) (ii) and (iii) (C) (i) and (ii) (D) (iii) and (iv)

ANSWER: (C) (i) and (ii)

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CHAPTER: 1 - Chemical Reactions and Equations

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: A metal, 'X', on treatment with sodium hydroxide liberates a gas 'G'. It also liberates the same gas 'G' on treatment with dilute sulphuric acid. Based on the above information, 'X' and 'G' respectively are:

OPTIONS: (A) Copper and Sulphur dioxide (B) Zinc and Sulphur dioxide (C) Zinc and Hydrogen (D) Copper and Hydrogen

ANSWER: (C) Zinc and Hydrogen

YEAR: 2025

CHAPTER: 1 - Chemical Reactions and Equations

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: A metal M displaces iron from aqueous solution of ferrous sulphate but fails to do so in case of aqueous solution of aluminium sulphate. The metal M is:

OPTIONS: (A) Magnesium (B) Copper (C) Lead (D) Zinc

ANSWER: (D) Zinc

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CHAPTER: 1 - Chemical Reactions and Equations

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The values of a, b, c and d in the following balanced chemical equation are respectively: $a \text{Pb(NO}_3)_2 \xrightarrow{\text{heat}} b \text{PbO} + c \text{NO}_2 + d \text{O}_2$

OPTIONS: (A) 1, 1, 2, 1 (B) 1, 1, 1, 2 (C) 2, 2, 1, 4 (D) 2, 2, 4, 1

ANSWER: (D) 2, 2, 4, 1

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CHAPTER: 1 - Chemical Reactions and Equations

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: In the electrolysis of water, if P and Q are the gases collected in the test tubes enclosing the electrodes R and S, then which option has the correct matching?

OPTIONS: (A) P-Oxygen gas, R-Anode and Q-Hydrogen gas, S-Anode (B) P-Hydrogen gas, R-Cathode and Q-Oxygen gas, S-Cathode (C) P-Oxygen gas, R-Anode and Q-Hydrogen gas, S-Cathode (D) P-Hydrogen gas, R-Anode and Q-Oxygen gas, S-Cathode

ANSWER: (C) P-Oxygen gas, R-Anode and Q-Hydrogen gas, S-Cathode

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CHAPTER: 1 - Chemical Reactions and Equations

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Select from the following a statement which is not true about burning of magnesium ribbon in air:

OPTIONS: (A) It burns with a dazzling white flame. (B) A white powder is formed on burning. (C) It is an endothermic reaction. (D) It is an example of a combination reaction.

ANSWER: (C) It is an endothermic reaction.

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CHAPTER: 1 - Chemical Reactions and Equations

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: If we make carbon skeleton with four carbon atoms, the two different possible skeletons will be:

OPTIONS: (A) Straight chain only (B) Branched chain only (C) Straight chain and branched chain (D) Ring structure and straight chain

ANSWER: (C) Straight chain and branched chain

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CHAPTER: 1 - Chemical Reactions and Equations

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 1

QUESTION: (i) $\text{AgNO}_3 + \text{NaCl} \rightarrow \text{NaNO}_3 + \text{AgCl}$ (ii) $\text{K}_2\text{SO}_4 + \text{BaCl}_2 \rightarrow \text{BaSO}_4 + 2\text{KCl}$. Which of the following options clearly describes both the reactions?

OPTIONS: (A) (i) is double displacement, (ii) is displacement reaction. (B) Both, (i) and (ii) are displacement reactions and precipitation reactions. (C) Both, (i) and (ii) are double displacement reactions and precipitation reactions. (D) (i) is displacement, (ii) is double displacement reaction.

ANSWER: (C) Both, (i) and (ii) are double displacement reactions and precipitation reactions.

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CHAPTER: 1 - Chemical Reactions and Equations

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 1

QUESTION: The gases evolved on heating lead (II) nitrate crystals are:

OPTIONS: (A) NO and O₂ (B) N₂ and NO₂ (C) NO₂ and H₂ (D) NO₂ and O₂

ANSWER: (D) NO₂ and O₂

YEAR: 2025

CHAPTER: 1 - Chemical Reactions and Equations

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 1

QUESTION: Which of the following is an example of a displacement reaction?

OPTIONS: (A) The combustion of natural gas (B) The reaction of a reactive metal with dilute acid (C) The electrolysis of water (D) The reaction of quick lime with water

ANSWER: (B) The reaction of a reactive metal with dilute acid

YEAR: 2025

CHAPTER: 1 - Chemical Reactions and Equations

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 1

QUESTION: To a small amount of copper oxide in a beaker, when we add dilute hydrochloric acid slowly with stirring, the change in the colour of the solution is due to the formation of:

OPTIONS: (A) Copper(II) oxide, which is black in colour. (B) Copper(I) chloride, which is blue in colour. (C) Copper(II) chloride, which is blue-green in colour. (D) Copper(I) oxide, which is black in colour.

ANSWER: (C) Copper(II) chloride, which is blue-green in colour.

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CHAPTER: 1 - Chemical Reactions and Equations

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 1

QUESTION: In the following equation $x\text{KI} + \text{H}_2\text{O}_2 \rightarrow \text{I}_2 + y\text{KOH}$, x and y respectively are:

OPTIONS: (A) 2,2 (B) 1,2 (C) 2,1 (D) 1,1

ANSWER: (A) 2,2

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CHAPTER: 1 - Chemical Reactions and Equations

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Alloys are homogeneous mixture of metal with a metal or non-metal.

Which among the following alloys contain non-metal as one of its constituents?

OPTIONS: (A) Brass (B) Bronze (C) Amalgam (D) Steel

ANSWER: (D) Steel

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CHAPTER: 1 - Chemical Reactions and Equations

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 1

QUESTION: Which of the following are decomposition reactions? (i) $2\text{KClO}_3 \rightarrow 2\text{KCl} + 3\text{O}_2$ (ii) $\text{MgO} + \text{H}_2\text{O} \rightarrow \text{Mg}(\text{OH})_2$ (iii) $2\text{H}_2\text{O} \rightarrow \text{H}_2 + \text{O}_2$ (iv) $\text{Zn} + \text{CuSO}_4 \rightarrow \text{ZnSO}_4 + \text{Cu}$

OPTIONS: (A) (i) and (ii) (B) (iii) and (iv) (C) (i) and (iii) (D) (ii) and (iv)

ANSWER: (C) (i) and (iii)

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CHAPTER: 2 - Acids, Bases and Salts (Chemistry)

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YEAR: 2025

CHAPTER: 2 - Acids, Bases and Salts

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The body of human beings works within the pH range of:

OPTIONS: (A) 6.1 to 6.8 (B) 6.5 to 7.3 (C) 7.0 to 7.8 (D) 7.5 to 8.1

ANSWER: (C) 7.0 to 7.8

YEAR: 2025

CHAPTER: 2 - Acids, Bases and Salts

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Which of the given option represents a family of salts?

OPTIONS: (A) NaCl, Na₂SO₄, CaSO₄ (B) K₂SO₄, Na₂SO₄, CaSO₄ (C) NaNO₃, CaCO₃, Na₂CO₃ (D) MgSO₄, CuSO₄, MgCl₂

ANSWER: (B) K₂SO₄, Na₂SO₄, CaSO₄

YEAR: 2025

CHAPTER: 2 - Acids, Bases and Salts

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: You have three aqueous solutions A, B and C as given below: A - Potassium nitrate, B - Ammonium chloride, C - Sodium carbonate. The ascending order of the pH of these solutions is:

OPTIONS: (A) A<B<C (B) B<C<A (C) C<A<B (D) B<A<C

ANSWER: (D) B<A<C

YEAR: 2025

CHAPTER: 2 - Acids, Bases and Salts

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The colour of the solution observed after about 1 hour of placing iron nails in copper sulphate solution is:

OPTIONS: (A) Blue (B) Pale green (C) Yellow (D) Reddish brown

ANSWER: (B) Pale green

YEAR: 2025

CHAPTER: 2 - Acids, Bases and Salts

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Juice of tamarind turns blue litmus to red. It is because of the presence of a chemical compound called:

OPTIONS: (A) Acetic acid (B) Methanoic acid (C) Oxalic acid (D) Tartaric acid

ANSWER: (D) Tartaric acid

YEAR: 2025

CHAPTER: 2 - Acids, Bases and Salts

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The water of crystallization is present in (i) Bleaching Powder (ii) Plaster of Paris (iii) Washing Soda (iv) Baking Soda:

OPTIONS: (A) (ii) and (iv) (B) (ii) and (iii) (C) (i) and (iii) (D) (i) and (iv)

ANSWER: (B) (ii) and (iii)

YEAR: 2025

CHAPTER: 2 - Acids, Bases and Salts

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: When a mixture of baking soda and tartaric acid is heated (or mixed in water) a product 'X' is formed, which is responsible for making breads and cakes soft and spongy. The product 'X' is:

OPTIONS: (A) Carbon dioxide (B) Carbon monoxide (C) Sodium tartrate (D) Hydrogen

ANSWER: (A) Carbon dioxide

YEAR: 2025

CHAPTER: 2 - Acids, Bases and Salts

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Which one of the following can be used as an acid-base indicator by a visually impaired (blind) student?

OPTIONS: (A) Turmeric (B) Vanilla essence (C) Methyl orange (D) Litmus

ANSWER: (B) Vanilla essence

YEAR: 2025

CHAPTER: 2 - Acids, Bases and Salts

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 2

QUESTION: How is tooth decay related to pH and how can it be prevented?

OPTIONS: (A) Tooth decay occurs when pH falls below 5.5 due to acid formation by bacteria; prevention includes brushing with fluoride toothpaste and avoiding sugary foods (B) Tooth decay occurs when pH rises above 8 due to alkaline conditions; prevention includes using acidic mouthwash (C) Tooth decay occurs when pH is exactly 7; prevention includes eating more sugar (D) Tooth decay occurs only due to physical damage; pH has no role

ANSWER: (A) Tooth decay occurs when pH falls below 5.5 due to acid formation by bacteria; prevention includes brushing with fluoride toothpaste and avoiding sugary foods

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CHAPTER: 3 - Metals and Non-Metals (Chemistry)

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YEAR: 2025

CHAPTER: 3 - Metals and Non-Metals

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Which of the following is a poor conductor of electricity?

OPTIONS: (A) Pb (B) Cu (C) Ag (D) Al

ANSWER: (A) Pb

YEAR: 2025

CHAPTER: 3 - Metals and Non-Metals

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Aluminium powder is used in thermit welding because:

OPTIONS: (A) Its reaction with iron is highly exothermic. (B) When it is heated with iron (III) oxide, molten iron is obtained. (C) When it is heated with iron (III) oxide, molten aluminium oxide is obtained to join railway tracks. (D) Its melting point is low as compared to iron and a molten alloy of iron and aluminium is formed on heating which is used to join railway tracks.

ANSWER: (B) When it is heated with iron (III) oxide, molten iron is obtained.

YEAR: 2025

CHAPTER: 3 - Metals and Non-Metals

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: A few pieces of granulated zinc are taken in a test tube and 2 mL of sodium hydroxide solution is added to it. When the contents are warmed, the product formed is:

OPTIONS: (A) Na_2ZnO (B) NaZnO_2 (C) Na_2ZnO_2 (D) NaZn(OH)_2

ANSWER: (C) Na_2ZnO_2

YEAR: 2025

CHAPTER: 3 - Metals and Non-Metals

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Two metals zinc and tin are dissolved separately in definite proportions in molten copper (the primary metal) to obtain two different alloys respectively known as:

OPTIONS: (A) Bronze and Brass (B) Brass and Solder (C) Brass and Bronze (D) Solder and Bronze

ANSWER: (C) Brass and Bronze

YEAR: 2025

CHAPTER: 3 - Metals and Non-Metals

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The most common method of extraction of metals from their oxide ores is:

OPTIONS: (A) Reduction with carbon (B) Reduction with hydrogen (C) Reduction with aluminium (D) Electrolytic reduction

ANSWER: (A) Reduction with carbon

YEAR: 2025

CHAPTER: 3 - Metals and Non-Metals

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Which of the following metals neither reacts with cold water nor reacts with hot water but reacts with steam?

OPTIONS: (A) Sodium (B) Calcium (C) Iron (D) Magnesium

ANSWER: (C) Iron

YEAR: 2025

CHAPTER: 3 - Metals and Non-Metals

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The metals obtained from their molten chlorides by the process of electrolytic reduction are:

OPTIONS: (A) Gold and silver (B) Calcium and magnesium (C) Aluminium and silver (D) Sodium and iron

ANSWER: (B) Calcium and magnesium

YEAR: 2025

CHAPTER: 3 - Metals and Non-Metals

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 3

QUESTION: What change is observed when a China dish containing copper powder is heated over the flame of a burner? What is the name of the phenomenon and the product formed?

OPTIONS: (A) Copper powder melts to form a liquid; it is a physical change; product is molten copper (B) Copper powder turns black due to formation of copper oxide; it is an oxidation reaction; the product is copper(II) oxide (C) Copper powder turns green due to formation of copper carbonate; it is a corrosion reaction; product is copper carbonate (D) Copper powder remains unchanged as copper does not react with air

ANSWER: (B) Copper powder turns black due to formation of copper oxide; it is an oxidation reaction; the product is copper(II) oxide

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CHAPTER: 4 - Carbon and its Compounds (Chemistry)

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YEAR: 2025

CHAPTER: 4 - Carbon and its Compounds

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: A Hydrocarbon which does not belong to the same homologous series of carbon compounds is:

OPTIONS: (A) C_4H_{10} (B) C_6H_{14} (C) C_7H_{14} (D) $C_{10}H_{22}$

ANSWER: (C) C_7H_{14}

YEAR: 2025

CHAPTER: 4 - Carbon and its Compounds

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Which of the following will not undergo addition reaction?

OPTIONS: (A) C_4H_8 (B) C_2H_2 (C) C_3H_8 (D) C_2H_4

ANSWER: (C) C_3H_8

YEAR: 2025

CHAPTER: 4 - Carbon and its Compounds

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 1

QUESTION: Which of the following set of compounds does not belong to same homologous series?

OPTIONS: (A) CH_4 and C_4H_{10} (B) C_2H_6 and C_3H_8 (C) C_3H_8 and C_5H_{12} (D) C_4H_8 and C_5H_{12}

ANSWER: (D) C_4H_8 and C_5H_{12}

YEAR: 2025

CHAPTER: 4 - Carbon and its Compounds

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 1

QUESTION: Which of the following is an amphoteric oxide?

OPTIONS: (A) Na₂O (B) K₂O (C) CO₂ (D) Al₂O₃

ANSWER: (D) Al₂O₃

YEAR: 2025

CHAPTER: 4 - Carbon and its Compounds

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 1

QUESTION: The formula of functional group for aldehyde is:

OPTIONS: (A) -COOH (B) -CHO (C) -C- (D) -OH

ANSWER: (B) -CHO

YEAR: 2025

CHAPTER: 4 - Carbon and its Compounds

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Which of the following compounds will undergo addition reaction?

OPTIONS: (A) C₄H₁₀ (B) C₄H₈ (C) C₃H₈ (D) CH₄

ANSWER: (B) C₄H₈

YEAR: 2025

CHAPTER: 4 - Carbon and its Compounds

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 5

QUESTION: A carbon compound P of molecular formula C₂H₆O is neutral to litmus and produces a gas that burns with a pop sound when reacted with sodium metal. What is compound P?

OPTIONS: (A) Methanol (B) Ethanol (C) Ethanoic acid (D) Methanoic acid

ANSWER: (B) Ethanol

YEAR: 2025

CHAPTER: 4 - Carbon and its Compounds

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 5

QUESTION: When ethanol is heated with concentrated H_2SO_4 at 443 K, what is the product formed and what is the role of H_2SO_4 ?

OPTIONS: (A) Ethene is formed; H_2SO_4 acts as a dehydrating agent (B) Ethanoic acid is formed; H_2SO_4 acts as an oxidizing agent (C) Ethane is formed; H_2SO_4 acts as a reducing agent (D) Ethyne is formed; H_2SO_4 acts as a catalyst

ANSWER: (A) Ethene is formed; H_2SO_4 acts as a dehydrating agent

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CHAPTER: 4 - Carbon and its Compounds

ORIGINAL_SECTION: B

ORIGINAL_MARKS: 5

QUESTION: What is observed when ethanol is heated with ethanoic acid in presence of an acid catalyst?

OPTIONS: (A) A sweet-smelling ester is formed (B) A foul-smelling gas is evolved (C) A precipitate is formed (D) No reaction occurs

ANSWER: (A) A sweet-smelling ester is formed

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CHAPTER: 5 - Life Processes (Biology)

YEAR: 2025

CHAPTER: 5 - Life Processes

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Which structure in a leaf is mainly responsible for gaseous exchange?

OPTIONS: (A) Xylem (B) Stomata (C) Phloem (D) Cuticle

ANSWER: (B) Stomata

YEAR: 2025

CHAPTER: 5 - Life Processes

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Identify the correct statement for Spirogyra, Leishmania and Hydra:

OPTIONS: (A) They reproduce sexually. (B) They are unicellular. (C) They are multicellular. (D) They reproduce asexually.

ANSWER: (D) They reproduce asexually.

YEAR: 2025

CHAPTER: 5 - Life Processes

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Human brain has various parts or regions that help in different actions, responses and coordination. From the following, identify the part responsible for precision of voluntary actions:

OPTIONS: (A) Cerebrum (B) Cerebellum (C) Medulla (D) Pons

ANSWER: (B) Cerebellum

YEAR: 2025

CHAPTER: 5 - Life Processes

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Pancreas secretes pancreatic juice which contain certain enzyme that helps in digestion of food. Choose the correct option from the following:

OPTIONS: (A) Trypsin digests emulsified fats and lipase digests proteins. (B) Trypsin digests proteins and lipase digests emulsified fats. (C) Trypsin and lipase both digest fats. (D) Trypsin digests proteins and lipase digests carbohydrates.

ANSWER: (B) Trypsin digests proteins and lipase digests emulsified fats.

YEAR: 2025

CHAPTER: 5 - Life Processes

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Sex is determined by different factors in various species. However, in human beings, it is determined genetically. Which amongst the following option(s) is/are correct for human beings? (i) Gamete carrying X chromosome from female parent. (ii)

Gamete carrying X chromosome from male parent. (iii) Gamete carrying Y chromosome from male parent.

OPTIONS: (A) (ii) and (iii) (B) (i) only (C) (i) and (iii) (D) (iii) only

ANSWER: (A) (ii) and (iii)

YEAR: 2025

CHAPTER: 5 - Life Processes

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Which of the following group is not 'biodegradable'?

OPTIONS: (A) Vegetable peels, dead leaves, paper (B) Cow dung, leather bag, water (C) Polythene bag, rubber band, ball pen (D) Paper, fruits, bones

ANSWER: (C) Polythene bag, rubber band, ball pen

YEAR: 2025

CHAPTER: 5 - Life Processes

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Which of the following limits the number of trophic levels in a food chain?

OPTIONS: (A) Availability of water (B) Deficient food supply (C) Decrease in energy at higher levels (D) Pollution in air

ANSWER: (C) Decrease in energy at higher levels

YEAR: 2025

CHAPTER: 5 - Life Processes

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: In a synapse, the chemical signal is transmitted from:

OPTIONS: (A) Dendritic end of one neuron to axonal end of another neuron. (B) Axon to cell body of the same neuron. (C) Cell body to axonal end of the same neuron. (D) Axonal end of one neuron to dendritic end of another neuron.

ANSWER: (D) Axonal end of one neuron to dendritic end of another neuron

YEAR: 2025

CHAPTER: 5 - Life Processes

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Organisms of a higher trophic level which feed on several types of organisms belonging to a lower trophic level form the:

OPTIONS: (A) Food web (B) Ecosystem (C) Ecological pyramid (D) Food chain

ANSWER: (A) Food web

YEAR: 2025

CHAPTER: 5 - Life Processes

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: In accumulation of toxic substances at higher trophic levels of an ecosystem through the food chain, which of the following organisms is affected the most?

OPTIONS: (A) Small aquatic plants (B) Human beings (C) Small fish (D) Large fish

ANSWER: (B) Human beings

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CHAPTER: 5 - Life Processes

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The growth of tendrils in grape plants is due to:

OPTIONS: (A) Effect of light (B) Effect of gravity (C) Rapid cell division in tendril cells in contact with the support (D) Rapid cell division in tendril cells that are away from the support

ANSWER: (D) Rapid cell division in tendril cells that are away from the support

YEAR: 2025

CHAPTER: 5 - Life Processes

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: In plants, the role of cytokinin is to:

OPTIONS: (A) promote cell division (B) cause wilting of leaves (C) promote opening of stomatal pore (D) promote growth of stem

ANSWER: (A) promote cell division

YEAR: 2025

CHAPTER: 5 - Life Processes

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The number of chromosomes in parents and offspring of a particular species undergoing sexual reproduction remains constant due to:

OPTIONS: (A) doubling of chromosomes after zygote formation. (B) halving of chromosomes after zygote formation. (C) doubling of chromosomes before gamete formation. (D) halving of chromosomes at the time of gamete formation.

ANSWER: (D) halving of chromosomes at the time of gamete formation.

YEAR: 2025

CHAPTER: 5 - Life Processes

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: One-cell thick blood vessels are known as:

OPTIONS: (A) Alveoli (B) Capillaries (C) Arteries (D) Veins

ANSWER: (B) Capillaries

YEAR: 2025

CHAPTER: 5 - Life Processes

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Select the correct option from the following statements about the functioning of the human heart:

OPTIONS: (A) Right atrium receives deoxygenated blood from different parts of the body and sends it to pulmonary veins. (B) Left atrium sends oxygenated blood to right ventricle which pumps it to different parts of the body. (C) Right atrium receives

deoxygenated blood from the body and sends it to the right ventricle. (D) Left atrium receives oxygenated blood from the pulmonary arteries and sends it to the left ventricle.
ANSWER: (C) Right atrium receives deoxygenated blood from the body and sends it to the right ventricle.

YEAR: 2025

CHAPTER: 5 - Life Processes

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Which of the following is the correct sequence of parts of female reproductive system of flowering plants in terms of their placement?

OPTIONS: (A) Stigma, ovule, ovary, style (B) Ovule, stigma, ovary, style (C) Style, stigma, ovule, ovary (D) Stigma, style, ovary, ovule

ANSWER: (D) Stigma, style, ovary, ovule

YEAR: 2025

CHAPTER: 5 - Life Processes

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The growth of the pollen tubes towards ovules is an example of:

OPTIONS: (A) Phototropism (B) Hydrotropism (C) Geotropism (D) Chemotropism

ANSWER: (D) Chemotropism

YEAR: 2025

CHAPTER: 5 - Life Processes

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: A pair of endocrine glands located in the human brain is:

OPTIONS: (A) Parathyroid and Pituitary (B) Pineal and Thymus (C) Hypothalamus and Thymus (D) Hypothalamus and Pineal

ANSWER: (D) Hypothalamus and Pineal

YEAR: 2025

CHAPTER: 5 - Life Processes

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The basic filtration unit of the excretory system in human beings is:

OPTIONS: (A) Nephron (B) Urethra (C) Neuron (D) Urinary bladder

ANSWER: (A) Nephron

YEAR: 2025

CHAPTER: 5 - Life Processes

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: In human alimentary canal, the digestive juice secreted by the gastric glands are:

OPTIONS: (A) Bile, Trypsin, Pepsin (B) Hydrochloric acid, Pepsin, Mucus (C) Lipase, Bile, Mucus (D) Salivary amylase, Pepsin, Bile

ANSWER: (B) Hydrochloric acid, Pepsin, Mucus

YEAR: 2025

CHAPTER: 5 - Life Processes

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Identify the part that controls the closing and opening of the stomatal pore in leaves of plants:

OPTIONS: (A) Stomata (B) Epidermal cells (C) Guard cells (D) Chloroplasts

ANSWER: (C) Guard cells

YEAR: 2025

CHAPTER: 5 - Life Processes

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Choose the equation of reaction that correctly represents anaerobic respiration in muscles:

OPTIONS: (A) Glucose → Pyruvate --Absence of O₂--> Ethanol + CO₂ + Energy (B) Glucose --O₂--> Pyruvate --O₂--> Energy (C) Glucose --O₂--> Pyruvate --Lack of Oxygen--> Ethanol + Energy (D) Glucose → Pyruvate --Lack of Oxygen--> Lactic acid + Energy

ANSWER: (D) Glucose → Pyruvate --Lack of Oxygen--> Lactic acid + Energy

YEAR: 2025

CHAPTER: 5 - Life Processes

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: When a seed germinates the root grows downwards and a small shoot grows upward. This shoot is known as:

OPTIONS: (A) radicle (B) stem (C) cotyledon (D) plumule

ANSWER: (D) plumule

YEAR: 2025

CHAPTER: 5 - Life Processes

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Select the correct option having correct matching of the organism given in Column I with the mode of reproduction in Column II: P Leishmania - 1. Regeneration, Q Spirogyra - 2. Multiple Fission, R Planaria - 3. Binary Fission, S Plasmodium - 4. Fragmentation, 5. Budding.

OPTIONS: (A) P-4, Q-2, R-1, S-3 (B) P-3, Q-4, R-5, S-2 (C) P-3, Q-4, R-1, S-2 (D) P-4, Q-3, R-2, S-1

ANSWER: (C) P-3, Q-4, R-1, S-2

YEAR: 2025

CHAPTER: 5 - Life Processes

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The plant hormones promoting rapid cell division in seeds and wilting of leaves respectively are:

OPTIONS: (A) Auxins and Absciscic acid (B) Cytokinins and Absciscic acid (C) Gibberellins and Auxins (D) Absciscic acid and Gibberellins
ANSWER: (B) Cytokinins and Absciscic acid

YEAR: 2025
CHAPTER: 5 - Life Processes
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: The modes of reproduction in Spirogyra and Planaria respectively are:
OPTIONS: (A) Regeneration and budding (B) Regeneration and fragmentation (C) Fragmentation and regeneration (D) Budding and regeneration
ANSWER: (C) Fragmentation and regeneration

YEAR: 2025
CHAPTER: 5 - Life Processes
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: The part of human eye which controls the amount of light entering into it:
OPTIONS: (A) Iris (B) Cornea (C) Ciliary muscles (D) Pupil
ANSWER: (A) Iris

YEAR: 2025
CHAPTER: 5 - Life Processes
ORIGINAL_SECTION: A
ORIGINAL_MARKS: 1
QUESTION: Mendel obtained F2 generation by:
OPTIONS: (A) self pollinating F1 generation plants (B) cross pollinating F1 generation plants with plants having dominant trait. (C) cross pollinating F1 generation plants with plants having recessive trait. (D) cross pollinating both the parents.
ANSWER: (A) self pollinating F1 generation plants

YEAR: 2025

CHAPTER: 5 - Life Processes

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: What did Mendel observe when he crossed pure tall pea plants with pure short pea plants?

OPTIONS: (A) All plants were short in F1 generation (B) All plants were tall in F1 generation (C) 75% plants were tall and 25% were short in F1 generation (D) 50% plants were tall and 50% were short in F1 generation

ANSWER: (B) All plants were tall in F1 generation

YEAR: 2025

CHAPTER: 5 - Life Processes

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 3

QUESTION: The gene combination of purple flowered pea plants is denoted as (WW) and that of white flowered pea plants as (ww). What was observed by Mendel in the F1 generation?

OPTIONS: (A) All plants had white flowers (B) All plants had purple flowers (C) 50% had purple and 50% had white flowers (D) 75% had purple and 25% had white flowers

ANSWER: (B) All plants had purple flowers

YEAR: 2025

CHAPTER: 5 - Life Processes

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 3

QUESTION: In the F2 generation from a cross between purple (WW) and white (ww) flowered pea plants, what is the percentage of white flowered plants?

OPTIONS: (A) 0% (B) 25% (C) 50% (D) 75%

ANSWER: (B) 25%

YEAR: 2025

CHAPTER: 5 - Life Processes

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 3

QUESTION: What is the ratio of gene combinations WW, Ww and ww in the F2 generation from a monohybrid cross?

OPTIONS: (A) 1:1:1 (B) 1:2:1 (C) 2:1:1 (D) 3:1:0

ANSWER: (B) 1:2:1

YEAR: 2025

CHAPTER: 5 - Life Processes

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 3

QUESTION: Which of the following correctly represents a dominant and a recessive trait?

OPTIONS: (A) Dominant trait is expressed only in homozygous condition while recessive trait is expressed in both homozygous and heterozygous conditions (B) Dominant trait is expressed in both homozygous and heterozygous conditions while recessive trait is expressed only in homozygous condition (C) Both dominant and recessive traits are expressed only in homozygous condition (D) Both dominant and recessive traits are expressed in heterozygous condition

ANSWER: (B) Dominant trait is expressed in both homozygous and heterozygous conditions while recessive trait is expressed only in homozygous condition

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CHAPTER: 6 - Control and Coordination in Animals and Plants (Biology)

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YEAR: 2025

CHAPTER: 6 - Control and Coordination in Animals and Plants

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Human brain has various parts or regions that help in different actions, responses and coordination. From the following, identify the part responsible for precision of voluntary actions:

OPTIONS: (A) Cerebrum (B) Cerebellum (C) Medulla (D) Pons

ANSWER: (B) Cerebellum

YEAR: 2025

CHAPTER: 6 - Control and Coordination in Animals and Plants

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: In a synapse, the chemical signal is transmitted from:

OPTIONS: (A) Dendritic end of one neuron to axonal end of another neuron. (B) Axon to cell body of the same neuron. (C) Cell body to axonal end of the same neuron. (D) Axonal end of one neuron to dendritic end of another neuron.

ANSWER: (D) Axonal end of one neuron to dendritic end of another neuron

YEAR: 2025

CHAPTER: 6 - Control and Coordination in Animals and Plants

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The growth of tendrils in grape plants is due to:

OPTIONS: (A) Effect of light (B) Effect of gravity (C) Rapid cell division in tendril cells in contact with the support (D) Rapid cell division in tendril cells that are away from the support

ANSWER: (D) Rapid cell division in tendril cells that are away from the support

YEAR: 2025

CHAPTER: 6 - Control and Coordination in Animals and Plants

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The plant hormone present in greater concentration in the areas of rapidly dividing cells is:

OPTIONS: (A) Auxin (B) Cytokinins (C) Gibberellins (D) Absciscic acid

ANSWER: (B) Cytokinins

YEAR: 2025

CHAPTER: 6 - Control and Coordination in Animals and Plants

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The part of the hind-brain controlling involuntary actions such as salivation and vomiting in humans is:

OPTIONS: (A) Cerebellum (B) Cerebrum (C) Pons (D) Medulla

ANSWER: (D) Medulla

YEAR: 2025

CHAPTER: 6 - Control and Coordination in Animals and Plants

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Electrical impulse travels in a neuron from:

OPTIONS: (A) Nerve ending -> Axon -> Cell body -> Dendrite (B) Dendrite -> Cell body -> Axon -> Nerve ending (C) Cell body -> Dendrite -> Axon -> Nerve ending (D)

Dendrite -> Axon -> Nerve ending -> Cell body

ANSWER: (B) Dendrite -> Cell body -> Axon -> Nerve ending

YEAR: 2025

CHAPTER: 6 - Control and Coordination in Animals and Plants

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Walking in a straight line and riding a bicycle are activities which are possible due to coordination of which part of the brain?

OPTIONS: (A) Fore-Brain Cerebrum (B) Mid-Brain Cerebellum (C) Hind-Brain Cerebellum (D) Hind-Brain Medulla

ANSWER: (C) Hind-Brain Cerebellum

YEAR: 2025

CHAPTER: 6 - Control and Coordination in Animals and Plants

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The part of the brain which maintains the posture and balance of the body is:

OPTIONS: (A) Pons (B) Cerebrum (C) Cerebellum (D) Medulla

ANSWER: (C) Cerebellum

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CHAPTER: 7 - How do Organisms Reproduce? (Biology)

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YEAR: 2025

CHAPTER: 7 - How do Organisms Reproduce?

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Identify the correct statement for Spirogyra, Leishmania and Hydra:

OPTIONS: (A) They reproduce sexually. (B) They are unicellular. (C) They are multicellular. (D) They reproduce asexually.

ANSWER: (D) They reproduce asexually.

YEAR: 2025

CHAPTER: 7 - How do Organisms Reproduce?

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Spirogyra, despite being a multi-cellular organism, still shows a simple reproductive method. Select the correct option with regard to reproduction in Spirogyra:

OPTIONS: (A) Asexual reproduction by division of cells into two cells (B) Sexual reproduction by division of cells into many cells (C) Asexual reproduction by breaking up of Spirogyra filaments into smaller parts (D) Sexual reproduction by the formation of gametes after meiosis of cells of filaments followed by fusion of male and female gametes

ANSWER: (C) Asexual reproduction by breaking up of Spirogyra filaments into smaller parts

YEAR: 2025

CHAPTER: 7 - How do Organisms Reproduce?

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The number of chromosomes in parents and offspring of a particular species undergoing sexual reproduction remains constant due to:

OPTIONS: (A) doubling of chromosomes after zygote formation. (B) halving of chromosomes after zygote formation. (C) doubling of chromosomes before gamete formation. (D) halving of chromosomes at the time of gamete formation.

ANSWER: (D) halving of chromosomes at the time of gamete formation.

YEAR: 2025

CHAPTER: 7 - How do Organisms Reproduce?

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Plants like rose and banana have lost the capacity to produce:

OPTIONS: (A) flowers (B) buds (C) seeds (D) fruits

ANSWER: (C) seeds

YEAR: 2025

CHAPTER: 7 - How do Organisms Reproduce?

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: In a bisexual flower the male gametes are present in the:

OPTIONS: (A) anther (B) ovary (C) stigma (D) filament

ANSWER: (A) anther

YEAR: 2025

CHAPTER: 7 - How do Organisms Reproduce?

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The correct/true statement(s) for a bisexual flower is/are: (i) They possess both stamen and pistil. (ii) They possess either stamen or pistil. (iii) They exhibit either self-pollination or cross-pollination. (iv) They cannot produce fruits on their own.

OPTIONS: (A) (i) only (B) (iv) only (C) (i) and (iii) (D) (i) and (iv)

ANSWER: (C) (i) and (iii)

YEAR: 2025

CHAPTER: 7 - How do Organisms Reproduce?

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Bryophyllum produces new plant through:

OPTIONS: (A) Apical buds formed on the tip of the plant (B) Vegetative buds produced in the notches of the leaf (C) Flowers produced in the notches of the branches (D) Fruits formed on the branches of the plant

ANSWER: (B) Vegetative buds produced in the notches of the leaf

YEAR: 2025

CHAPTER: 7 - How do Organisms Reproduce?

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The number of chromosomes in a cell division is halved. This kind of cell division is observed in:

OPTIONS: (A) Only testis (B) Only ovary (C) Ovary and testis both (D) All cells of the body

ANSWER: (C) Ovary and testis both

YEAR: 2025

CHAPTER: 7 - How do Organisms Reproduce?

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Which one of the following has half the number of chromosomes and half the amount of DNA as compared to the non-reproductive body cells?

OPTIONS: (A) Male germ cell (B) Female germ cell (C) Zygote (D) Both, male and female germ cells

ANSWER: (D) Both, male and female germ cells

YEAR: 2025

CHAPTER: 7 - How do Organisms Reproduce?

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Match Column- I with Column- II and select the correct option from the choices provided. a. Site of fertilisation of egg with the sperm - (i) Vagina, b. Site of implantation of embryo - (ii) Uterus, c. Site of entry of sperm into the female reproductive tract - (iii) Oviduct, d. Site through which the waste materials generated by the developing embryo are removed - (iv) Placenta, (v) Cervix.

OPTIONS: (A) a- (ii), b- (i), c- (v), d- (iv) (B) a- (iii), b- (i), c- (v), d- (iv) (C) a- (iv), b- (ii), c- (iii), d- (i) (D) a- (iii), b- (ii), c- (i), d- (iv)

ANSWER: (D) a- (iii), b- (ii), c- (i), d- (iv)

YEAR: 2025

CHAPTER: 7 - How do Organisms Reproduce?

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Select a pair of bisexual flowers from the following:

OPTIONS: (A) Papaya and mustard (B) Hibiscus and mustard (C) Hibiscus and papaya (D) Hibiscus and watermelon

ANSWER: (B) Hibiscus and mustard

YEAR: 2025

CHAPTER: 7 - How do Organisms Reproduce?

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Listed below are the steps of nutrition in Amoeba. Select the correct sequence of these steps: (i) Diffusion of simple nutrients into cytoplasm (ii) Food vacuole formation (iii) Formation of finger like temporary extensions of cell surface (iv) Complex substances broken to simpler ones (v) Undigested material thrown out of the cell surface.

OPTIONS: (A) (iv), (i), (ii), (iii), (v) (B) (iii), (ii), (iv), (i), (v) (C) (ii), (i), (iv), (v), (iii) (D) (iii), (iv), (i), (ii), (v)

ANSWER: (B) (iii), (ii), (iv), (i), (v)

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CHAPTER: 8 - Heredity and Evolution (Biology)

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YEAR: 2025

CHAPTER: 8 - Heredity and Evolution

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The genotype of green stemmed tomato plant is denoted by 'GG' and that of purple stemmed tomato plant as 'gg'. Hybrid of a cross between these two were self-pollinated to obtain F₂ progeny. What will be the ratio of GG, Gg and gg in this F₂ progeny?

OPTIONS: (A) 2:1:1 (B) 3:1:0 (C) 1:1:2 (D) 1:2:1

ANSWER: (D) 1:2:1

YEAR: 2025

CHAPTER: 8 - Heredity and Evolution

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: If pea plants with round and green seeds (RRyy) are crossed with pea plants having wrinkled and yellow seeds (rrYY), the seeds developed by the plants of F₁ generation will be:

OPTIONS: (A) 50% round and green (B) 75% wrinkled and green (C) 100% round and yellow (D) 75% wrinkled and yellow

ANSWER: (C) 100% round and yellow

YEAR: 2025

CHAPTER: 8 - Heredity and Evolution

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 3

QUESTION: When a pure-tall pea plant is crossed with a pure-dwarf pea plant, the percentage of tall pea plants in F₁ and F₂ generation pea plants will be respectively:

OPTIONS: (A) 100% ; 25% (B) 100% ; 50% (C) 100% ; 75% (D) 100% ; 100%

ANSWER: (C) 100% ; 75%

YEAR: 2025

CHAPTER: 8 - Heredity and Evolution

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: A pure pea plant bearing terminal flowers was cross-pollinated with a pure plant having axillary flowers. What will be the phenotype of F1 generation?

OPTIONS: (A) All terminal flowers (B) All axillary flowers (C) 50% terminal and 50% axillary flowers (D) 75% axillary and 25% terminal flowers

ANSWER: (B) All axillary flowers

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CHAPTER: 9 - Light - Reflection and Refraction (Physics)

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YEAR: 2025

CHAPTER: 9 - Light - Reflection and Refraction

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 1

QUESTION: A convex lens of focal length 15 cm is forming a real image. If the size of image is same as the size of object, then position of object and position of image will be, respectively:

OPTIONS: (A) -15 cm and -15 cm from lens (B) -15 cm and +15 cm from lens (C) -30 cm and +30 cm from lens (D) -30 cm and -30 cm from lens

ANSWER: (C) -30 cm and +30 cm from lens

YEAR: 2025

CHAPTER: 9 - Light - Reflection and Refraction

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 1

QUESTION: When you look at an object very close to your eyes, the:

OPTIONS: (A) Ciliary muscles of your eye contract and the eye lens becomes thick. (B) Ciliary muscles of your eye get relaxed and the eye lens becomes thick. (C) Ciliary muscles of your eye contract and the eye lens becomes thin. (D) Ciliary muscles of your eye get relaxed and the eye lens becomes thin.

ANSWER: (A) Ciliary muscles of your eye contract and the eye lens becomes thick.

YEAR: 2025

CHAPTER: 9 - Light - Reflection and Refraction

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 1

QUESTION: From the following options choose the one which gives negative magnification greater than one:

OPTIONS: (A) Concave mirror; Nature of image - real (B) Concave mirror; Nature of image - virtual (C) Convex mirror; Nature of image - real (D) Convex mirror; Nature of image - virtual

ANSWER: (A) Concave mirror; Nature of image - real

YEAR: 2025

CHAPTER: 9 - Light - Reflection and Refraction

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 1

QUESTION: A thin beam of white light falls on a face of a glass prism and a spectrum is obtained on a screen. Which of the following options is the correct sequence of colours from top to bottom on the screen?

OPTIONS: (A) Red, Green, Yellow, Violet, Blue (B) Red, Yellow, Green, Blue, Violet (C) Violet, Blue, Green, Yellow, Red (D) Violet, Red, Green, Yellow, Blue

ANSWER: (B) Red, Yellow, Green, Blue, Violet

YEAR: 2025

CHAPTER: 9 - Light - Reflection and Refraction

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 1

QUESTION: A candle flame is placed in front of the reflecting surface of a convex mirror of focal length f . If the distance of the flame from the pole of the mirror is ' f ', its image is formed:

OPTIONS: (A) at infinite distance from the mirror (B) behind the mirror at the principal focus (C) behind the mirror at a distance $2f$ (D) behind the mirror at a distance $f/2$

ANSWER: (D) behind the mirror at a distance $f/2$

YEAR: 2025

CHAPTER: 9 - Light - Reflection and Refraction

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 1

QUESTION: To get an image of magnification -1 on a screen using a lens of focal length 20 cm, the object distance must be:

OPTIONS: (A) Less than 20 cm (B) 30 cm (C) 40 cm (D) 80 cm

ANSWER: (C) 40 cm

YEAR: 2025

CHAPTER: 9 - Light - Reflection and Refraction

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 1

QUESTION: An optical device 'X' is placed obliquely in the path of a narrow parallel beam of light. If the emergent beam gets displaced laterally, the device 'X' is:

OPTIONS: (A) plane mirror (B) convex lens (C) glass slab (D) glass prism

ANSWER: (C) glass slab

YEAR: 2025

CHAPTER: 9 - Light - Reflection and Refraction

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: A person is suffering from an eye defect in which the far point of the eye is much nearer than infinity. What is the defect?

OPTIONS: (A) Hypermetropia (B) Myopia (C) Presbyopia (D) Cataract

ANSWER: (B) Myopia

YEAR: 2025

CHAPTER: 9 - Light - Reflection and Refraction

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: Which of the following is a cause of myopia?

OPTIONS: (A) Elongation of eyeball (B) Shortening of eyeball (C) Weakening of ciliary muscles (D) Loss of flexibility of eye lens

ANSWER: (A) Elongation of eyeball

YEAR: 2025

CHAPTER: 9 - Light - Reflection and Refraction

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: Which type of lens is used to correct myopia?

OPTIONS: (A) Convex lens (B) Concave lens (C) Bifocal lens (D) Cylindrical lens

ANSWER: (B) Concave lens

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CHAPTER: 10 - The Human Eye and the Colourful World (Physics)

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YEAR: 2025

CHAPTER: 10 - The Human Eye and the Colourful World

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 1

QUESTION: When you look at an object very close to your eyes, the:

OPTIONS: (A) Ciliary muscles of your eye contract and the eye lens becomes thick. (B) Ciliary muscles of your eye get relaxed and the eye lens becomes thick. (C) Ciliary muscles of your eye contract and the eye lens becomes thin. (D) Ciliary muscles of your eye get relaxed and the eye lens becomes thin.

ANSWER: (A) Ciliary muscles of your eye contract and the eye lens becomes thick.

YEAR: 2025

CHAPTER: 10 - The Human Eye and the Colourful World

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 1

QUESTION: The vision defect which arises due to gradual weakening of the ciliary muscles and diminishing flexibility of eye lens is:

OPTIONS: (A) Myopia (B) Hypermetropia (C) Presbyopia (D) Myopia and hypermetropia both

ANSWER: (C) Presbyopia

YEAR: 2025

CHAPTER: 10 - The Human Eye and the Colourful World

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 1

QUESTION: The pair of eye parts responsible for admitting different amount of light in to the eyes is:

OPTIONS: (A) Iris and pupil (B) Ciliary muscles and pupil (C) Retina and Iris (D) Ciliary muscles and cornea

ANSWER: (A) Iris and pupil

YEAR: 2025

CHAPTER: 10 - The Human Eye and the Colourful World

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 1

QUESTION: Select the correct statement from the following:

OPTIONS: (A) The size of the molecules of air is larger than the wavelength of visible light. (B) The blue light has a wavelength about 1.8 times greater than that of red light. (C) When sunlight passes through the fine particles in air, they scatter the blue colour of visible light more strongly than red. (D) The light of red colour is scattered the most by fog or smoke.

ANSWER: (C) When sunlight passes through the fine particles in air, they scatter the blue colour of visible light more strongly than red.

YEAR: 2025

CHAPTER: 10 - The Human Eye and the Colourful World

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 2

QUESTION: How does the focal length of the eye lens change as the distance of the object from the eyes is altered?

OPTIONS: (A) Focal length increases when viewing distant objects and decreases when viewing near objects (B) Focal length decreases when viewing distant objects and increases when viewing near objects (C) Focal length remains constant for all distances (D) Focal length changes only when viewing objects beyond 25 cm

ANSWER: (A) Focal length increases when viewing distant objects and decreases when viewing near objects

YEAR: 2025

CHAPTER: 10 - The Human Eye and the Colourful World

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 2

QUESTION: Why is the concave lens used as a corrective lens for a myopic eye?

OPTIONS: (A) It diverges light rays so that they focus on the retina (B) It converges light rays so that they focus on the retina (C) It changes the shape of the eye lens (D) It increases the size of the image

ANSWER: (A) It diverges light rays so that they focus on the retina

YEAR: 2025

CHAPTER: 10 - The Human Eye and the Colourful World

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: The power of a lens is -0.25 D. What is the type of lens and its focal length?

OPTIONS: (A) Convex lens, focal length $+4$ m (B) Concave lens, focal length -4 m (C) Convex lens, focal length $+0.25$ m (D) Concave lens, focal length -0.25 m

ANSWER: (B) Concave lens, focal length -4 m

YEAR: 2025

CHAPTER: 10 - The Human Eye and the Colourful World

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: A person has to keep reading material much beyond 25 cm (say at 50 cm) for comfortable reading. Name the defect of vision he is suffering from.

OPTIONS: (A) Myopia (B) Hypermetropia (C) Presbyopia (D) Cataract

ANSWER: (B) Hypermetropia

YEAR: 2025

CHAPTER: 10 - The Human Eye and the Colourful World

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: Which type of lens is used to correct hypermetropia?

OPTIONS: (A) Concave lens (B) Convex lens (C) Bifocal lens (D) Cylindrical lens

ANSWER: (B) Convex lens

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CHAPTER: 11 - Electricity (Physics)

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YEAR: 2025

CHAPTER: 11 - Electricity

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 1

QUESTION: A piece of wire of resistance 'R' is cut lengthwise into three identical parts. These parts are then connected in parallel. If the equivalent resistance of this combination is R', then the value of R/R' is:

OPTIONS: (A) 1/9 (B) 1/3 (C) 3 (D) 9

ANSWER: (D) 9

YEAR: 2025

CHAPTER: 11 - Electricity

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 1

QUESTION: An electric bulb is rated 220 V; 11 W. The resistance of its filament when it glows with a power supply of 220 V is:

OPTIONS: (A) 4400 Ω (B) 440 Ω (C) 400 Ω (D) 20 Ω

ANSWER: (A) 4400 Ω

YEAR: 2025

CHAPTER: 11 - Electricity

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 1

QUESTION: The minimum number of identical bulbs of rating 4 V; 6 W, that can work safely with desired brightness, when connected in series with a 240 V mains supply is:

OPTIONS: (A) 20 (B) 40 (C) 60 (D) 80

ANSWER: (C) 60

YEAR: 2025

CHAPTER: 11 - Electricity

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: State Ohm's law. Which of the following correctly represents it?

OPTIONS: (A) $V = IR$, where V is directly proportional to I at constant temperature (B) $V = R/I$, where V is inversely proportional to I (C) $I = VR$, where I is directly proportional to R (D) $V = I/R$, where V is inversely proportional to R

ANSWER: (A) $V = IR$, where V is directly proportional to I at constant temperature

YEAR: 2025

CHAPTER: 11 - Electricity

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: The V - I graph for Ohm's law will always be:

OPTIONS: (A) A straight line passing through the origin (B) A curved line passing through the origin (C) A straight line not passing through the origin (D) A horizontal line

ANSWER: (A) A straight line passing through the origin

YEAR: 2025

CHAPTER: 11 - Electricity

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 2

QUESTION: Define one ampere.

OPTIONS: (A) One ampere is the current flowing when one coulomb of charge passes through a point in one second (B) One ampere is the current flowing when one joule of energy is consumed per second (C) One ampere is the current flowing when one volt of

potential difference is applied across one ohm (D) One ampere is the current flowing when one watt of power is consumed

ANSWER: (A) One ampere is the current flowing when one coulomb of charge passes through a point in one second

YEAR: 2025

CHAPTER: 11 - Electricity

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: An object is placed at a distance of 30 cm from the pole of a concave mirror. If its real and inverted image is formed at 60 cm in front of the mirror, the focal length of the mirror is:

OPTIONS: (A) -15 cm (B) -20 cm (C) +20 cm (D) +15 cm

ANSWER: (B) -20 cm

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CHAPTER: 12 - Magnetic Effects of Electric Current (Physics)

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YEAR: 2025

CHAPTER: 12 - Magnetic Effects of Electric Current

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 1

QUESTION: Which one of the following statements is not true about a bar magnet?

OPTIONS: (A) It sets itself in north-south direction when suspended freely. (B) It has attractive power for iron filings. (C) It produces magnetic field lines. (D) The direction of magnetic field lines inside a bar magnet is from its north pole to its south pole.

ANSWER: (D) The direction of magnetic field lines inside a bar magnet is from its north pole to its south pole.

YEAR: 2025

CHAPTER: 12 - Magnetic Effects of Electric Current

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 1

QUESTION: The strength of magnetic field produced inside a long straight current carrying solenoid does not depend upon:

OPTIONS: (A) number of turns in the solenoid (B) direction of current flowing through the solenoid (C) material of the core filled inside the solenoid (D) radius of the coil of the solenoid

ANSWER: (D) radius of the coil of the solenoid

YEAR: 2025

CHAPTER: 12 - Magnetic Effects of Electric Current

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 1

QUESTION: In domestic electric circuits, the colour of insulation covers of wires in the cables of electric iron/electric toaster is generally:

OPTIONS: (A) red for live wire, green for neutral wire and black for earth wire (B) red for live wire, black for neutral wire and green for earth wire (C) green for live wire, black for neutral wire and red for earth wire (D) green for live wire, red for neutral wire and black for earth wire

ANSWER: (B) red for live wire, black for neutral wire and green for earth wire

YEAR: 2025

CHAPTER: 12 - Magnetic Effects of Electric Current

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: A current carrying conductor placed in an external magnetic field experiences a force. Which rule determines the direction of this force?

OPTIONS: (A) Right hand thumb rule (B) Fleming's left hand rule (C) Fleming's right hand rule (D) Left hand thumb rule

ANSWER: (B) Fleming's left hand rule

YEAR: 2025

CHAPTER: 12 - Magnetic Effects of Electric Current

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 3

QUESTION: An electron beam moving horizontally towards the front wall from the back wall is deflected to the right side by a magnetic field. The direction of the magnetic field is:

OPTIONS: (A) Upward (B) Downward (C) Towards the left wall (D) Towards the right wall

ANSWER: (A) Upward

YEAR: 2025

CHAPTER: 12 - Magnetic Effects of Electric Current

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 5

QUESTION: What is a fuse and why is it called a safety device?

OPTIONS: (A) It is a device that increases voltage; called safety device because it boosts power (B) It is a wire that melts and breaks the circuit when current exceeds safe limit; called safety device because it prevents damage due to overloading (C) It is a device that stores charge; called safety device because it stabilizes current (D) It is a switch that turns off manually; called safety device because it controls power

ANSWER: (B) It is a wire that melts and breaks the circuit when current exceeds safe limit; called safety device because it prevents damage due to overloading

YEAR: 2025

CHAPTER: 12 - Magnetic Effects of Electric Current

ORIGINAL_SECTION: C

ORIGINAL_MARKS: 5

QUESTION: Why are electrical appliances with metallic bodies connected to the mains through a three-pin plug, whereas an electric bulb is connected with a two-pin plug?

OPTIONS: (A) Metallic appliances need earth connection for safety in case of leakage; bulb does not need earth as it has no metallic body (B) Three-pin plug provides higher voltage; bulb needs lower voltage (C) Metallic appliances consume more power; bulb consumes less power (D) Three-pin plug is cheaper than two-pin plug

ANSWER: (A) Metallic appliances need earth connection for safety in case of leakage; bulb does not need earth as it has no metallic body

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CHAPTER: 13 - Our Environment (Biology)

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YEAR: 2025

CHAPTER: 13 - Our Environment

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Which of the following group is not 'biodegradable'?

OPTIONS: (A) Vegetable peels, dead leaves, paper (B) Cow dung, leather bag, water
(C) Polythene bag, rubber band, ball pen (D) Paper, fruits, bones

ANSWER: (C) Polythene bag, rubber band, ball pen

YEAR: 2025

CHAPTER: 13 - Our Environment

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Which of the following limits the number of trophic levels in a food chain?

OPTIONS: (A) Availability of water (B) Deficient food supply (C) Decrease in energy at
higher levels (D) Pollution in air

ANSWER: (C) Decrease in energy at higher levels

YEAR: 2025

CHAPTER: 13 - Our Environment

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Which one of the following gets biomagnified at different levels in a food
chain?

OPTIONS: (A) Carbon monoxide (B) CFC's (C) DDT (D) Manure

ANSWER: (C) DDT

YEAR: 2025

CHAPTER: 13 - Our Environment

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: In the food chains given below, select the most efficient food chain in terms of energy:

OPTIONS: (A) Grass -> Grasshopper -> Frog -> Snake (B) Plants -> Deer -> Lion (C) Plants -> Man (D) Phytoplankton -> Zooplankton -> Small Fish -> Big Fish

ANSWER: (C) Plants -> Man

YEAR: 2025

CHAPTER: 13 - Our Environment

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Consider the following food chain: Grass -> Grasshopper -> Frog -> Snake -> Eagle. If the amount of energy available at third trophic level is 50 kJ, the available energy at the producer level was:

OPTIONS: (A) 0.5 kJ (B) 5 kJ (C) 500 kJ (D) 5000 kJ

ANSWER: (D) 5000 kJ

YEAR: 2025

CHAPTER: 13 - Our Environment

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The incorrect statement about ozone is:

OPTIONS: (A) It is a deadly poisonous gas. (B) It shields the surface of the earth from UV radiation from sun. (C) It is used as a refrigerant and in fire-extinguishers. (D) It is formed by combining oxygen molecule with free oxygen atom.

ANSWER: (C) It is used as a refrigerant and in fire-extinguishers.

YEAR: 2025

CHAPTER: 13 - Our Environment

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Green plants occupy the first trophic level in every food chain because they:

OPTIONS: (A) exist over a large area. (B) have very less concentration of harmful chemicals. (C) have to feed large number of herbivores. (D) can synthesize food by photosynthesis.

ANSWER: (D) can synthesize food by photosynthesis.

YEAR: 2025

CHAPTER: 13 - Our Environment

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: In the following groups of wastes, which group contains only non-biodegradable wastes?

OPTIONS: (A) Leather footwear, Plastic plate, Polythene bag (B) Empty medicine bottle, Milk packet, Aluminium can (C) Used tea leaves, Cardboard box, Iron nail (D) Plastic Syringes, Newspaper, Ball point pen

ANSWER: (B) Empty medicine bottle, Milk packet, Aluminium can

YEAR: 2025

CHAPTER: 13 - Our Environment

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Which one of the following is not an excretory product in plants?

OPTIONS: (A) CO₂ (B) Starch (C) Resins and gums (D) Dead cells

ANSWER: (B) Starch

YEAR: 2025

CHAPTER: 13 - Our Environment

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: Human activities that are affecting the environment are:

OPTIONS: (A) minimising the use of chlorofluorocarbons. (B) excessive use of disposable cups and plates. (C) maximising the use of reusable utensils for eating food and drinking fluids. (D) segregating the wastes into biodegradable and non-biodegradable before disposal.

ANSWER: (B) excessive use of disposable cups and plates.

YEAR: 2025

CHAPTER: 13 - Our Environment

ORIGINAL_SECTION: A

ORIGINAL_MARKS: 1

QUESTION: The examples of natural and manmade (artificial) ecosystems are respectively:

OPTIONS: (A) Forests and ponds (B) Crop fields and lakes (C) Lakes and gardens (D) Crop fields and forests

ANSWER: (C) Lakes and gardens
